Intelligence

Lecture 6

Neuroscience of Intelligence

Dr Steven Poulter

Key Themes

- Brain structure and intelligence
- Brain cells and intelligence
- Brain regions and intelligence
- Brain connectivity and intelligence

Glossary of Key Terms

Brain Structure Terms

Grey Matter / Cortical Thickness
Thickness of the brain's outer layer containing neuron cell bodies. Thicker cortex correlates positively with *g* and IQ.

White Matter Integrity
Quality of axonal connections ("wires") enabling communication between regions. Higher integrity → higher *g*.

White Matter Hyperintensities / "Scars"
Age-related lesions interrupting processing efficiency. Higher WMH → lower intelligence.

Total Brain Volume
Volume inside skull minus cerebrospinal fluid. Meta-analytic correlations with intelligence ≈ .28–.31.

Glossary of Key Terms

Fluid Intelligence (*g*f)
Ability to solve novel problems, reason abstractly, and manage interference (e.g., Raven's Matrices). Linked strongly to working memory and PFC activity.

Crystallised Intelligence (gc)
Knowledge accumulated through experience (e.g., vocabulary, facts). Distinct from *g*f.

MRI (Magnetic Resonance Imaging)
Measures brain structure (grey matter thickness, white matter integrity, total brain volume).

fMRI (Functional MRI)
Measures blood flow changes during cognitive tasks. Used in Gray et al. (2003) to link working memory load to PFC activity.

EEG (Electroencephalography)
Measures electrical activity at rest or in response to stimuli.

MEG (Magnetoencephalography)
Detects magnetic fields created by neural electrical activity.

Glossary of Key Terms

Neuron / Brain Cell
Basic information-processing units of the brain.

Dendrites / Dendritic Complexity
Branch-like structures that collect inputs. Larger/more complex dendritic trees predict higher IQ.

Action Potential (AP)
Electrical signal fired by a neuron. Faster APs were observed in individuals with higher IQ.

Pyramidal Cells
Major excitatory neurons in the cortex; crucial for advanced cognition. Differences in dendritic size and AP speed correlate with intelligence.

Prefrontal Cortex (PFC)
Region linked to working memory, reasoning, interference control. Left lateral PFC activation correlates strongly with fluid intelligence.

Temporal Cortex
Region where cortical thickness correlates with IQ and where neuron samples were taken in Goriounova et al. (2018).

Working Memory (WM)
Short-term storage and manipulation of information. Strongly associated with *g*f. Assessed via the “3-back” task with lures.

n-Back Task
A cognitive task requiring recall of stimuli presented “n” steps earlier. Used to tax working memory/interference control.

Lure Trials
High-interference trials in the n-back task. Accuracy on lure trials correlates strongly with *g*f.

Connectome
Map of all neural connections in the brain.

Morphometric Similarity Networks (MSNs)
MRI-derived measures showing how similar brain regions are in structure, used to estimate connectivity.

Nodal Degree / Hubness
Number of connections a brain region has. Higher degree (“more hub-like”) regions in frontal and temporal cortex predict higher IQ.

Intelligence as an Orchestra?

London Philharmonic Orchestra

Local School Orchestra

Intelligence as an Orchestra?

Variation in size of orchestras and the quality of music performed by orchestras of the same size.

Intelligence as an Orchestra?

Studying orchestral quality by removing one player at a time - similar to studies looking at the effect of brain injury on different parts of the brain

Intelligence as an Orchestra?

Brass

Stringed

Woodwind

Percussion

Ways to measure the brain

Brain structure (MRI)

> different types of brain tissue a person has

> can indicate how healthy brain tissue is

Brain function (fMRI)

> blood flow changes during mental tasks

Electroencephalographic (EEG, or brain waves) and brain evoked response methods

> electrical activity at rest / in response to stimuli

Ways to measure the brain

Magnetoencephalography (MEG) methods

> magnetic activity from electrical activity of brain

Brain structure and intelligence

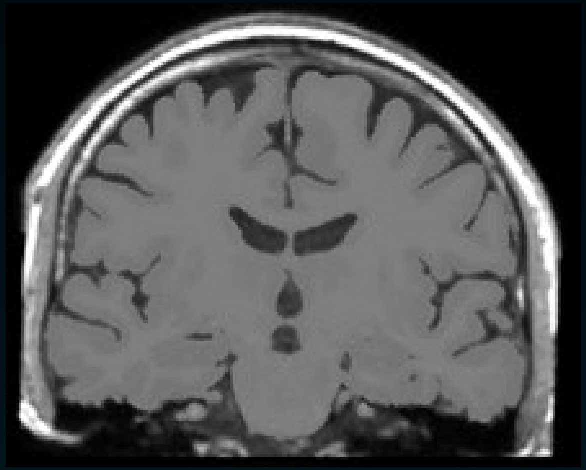
What do more intelligent brains look like?

- 672 people around the age of 73 years
- 15 cognitive tests – many of them from WAIS
- General intelligence (*g*) score calculated
- MRI scans to measure to measure different types of *structure* in the brain

Ritchie et al. (2015) *Intelligence*

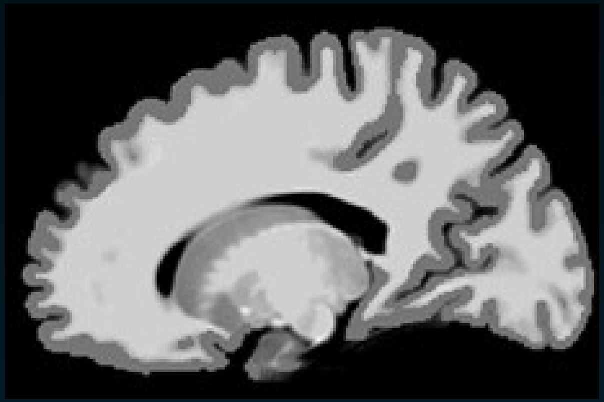
What do more intelligent brains look like?

Brain measurements taken from participants



Total brain volume
Total volume inside skull minus cerebrospinal fluid. How much brain, of any type of tissue, inside skull

Brain cortical thickness
thickness of the brains' grey matter – largely the neurons of the brain (info. processors)



Ritchie et al. (2015) *Intelligence*

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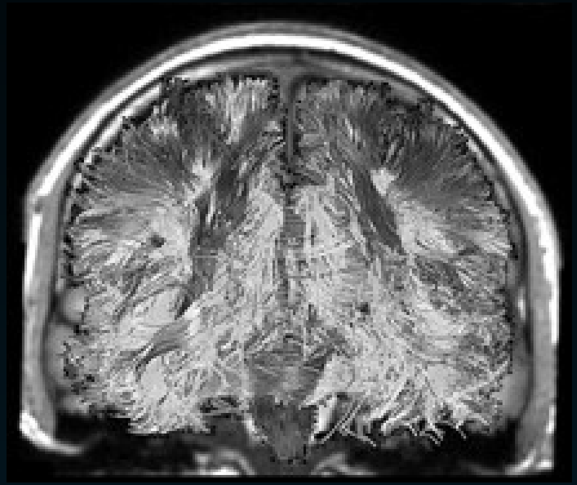
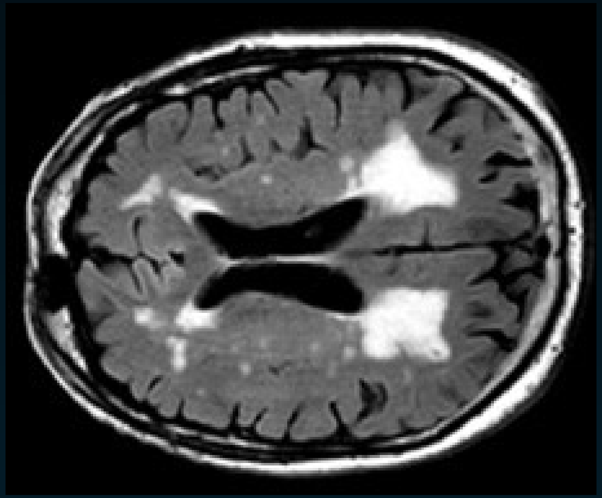


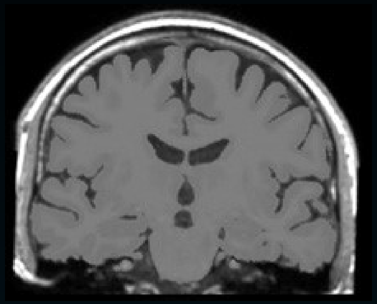
Brain white matter integrity
Connecting network of 'wires' in the brain – the axons connecting nerve cells to one another. Lies mostly below grey matter of the cortex

Brain white matter intensities
scars in the brain's connections. More develop as we age. Could interrupt efficient processing



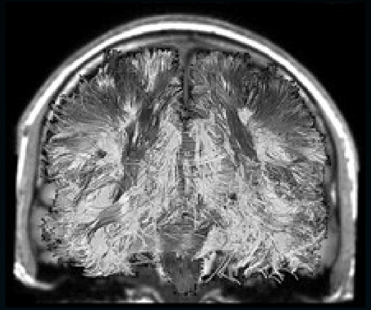
Ritchie et al. (2015) *Intelligence*

Brain measurements and correlation with *g*



Total brain volume

0.31

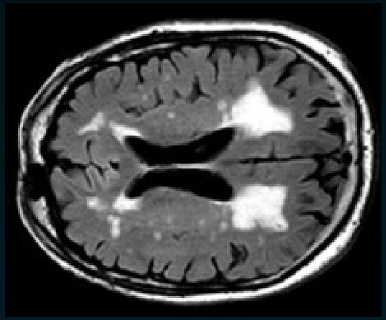


Brain white matter integrity

0.24

Brain cortical thickness

0.24



Brain white matter intensities

-0.20

Ritchie et al. (2015) *Intelligence*

Intellectual ability and cortical development in children and adolescents

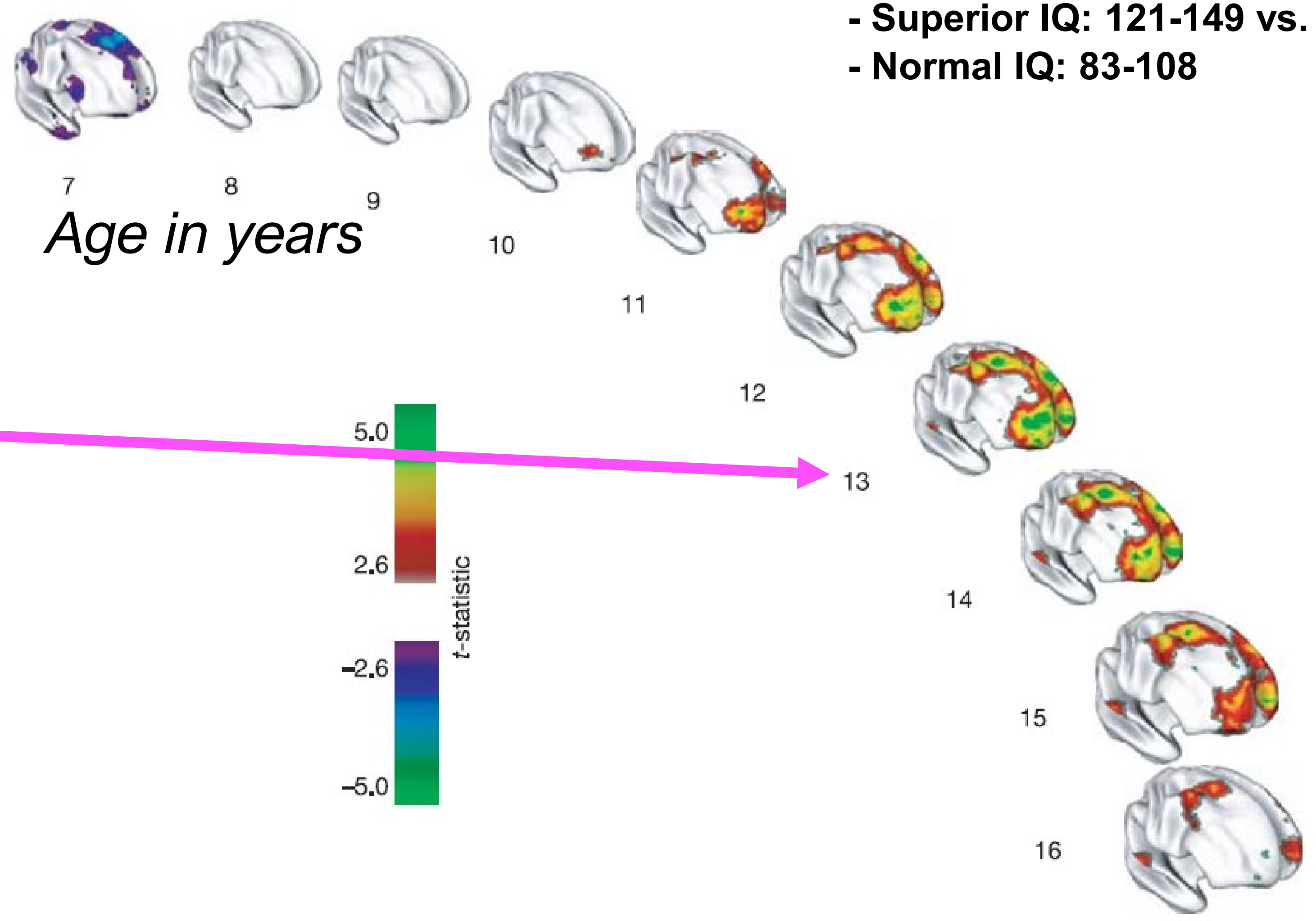
Nature (2006)

P. Shaw[1], D. Greenstein[1], J. Lerch[2], L. Clasen[1], R. Lenroot[1], N. Gogtay[1], A. Evans[2], J. Rapoport[1] & J. Giedd[1]

- Superior IQ: 121-149 vs.
- Normal IQ: 83-108

Superior IQ group has a thinner superior prefrontal cortex at the earliest age (purple regions)

Rapid increase in cortical thickness (red, green and yellow regions) in the superior IQ group (peaking at age 13, waning in late adolescence)



Dynamic neuroanatomical expression of intelligence

Brain volume and intelligence: The moderating role of intelligence measurement quality

Gilles E. Gignac[a,*], Timothy C. Bates[b]

Intelligence (2017)

- Found 32 correlations with a total of 1758 participants (healthy adults)
- Overall correlation of **0.29** between volume and general intelligence

Intelligence (2019)

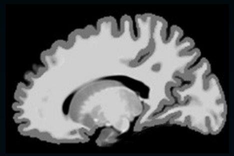
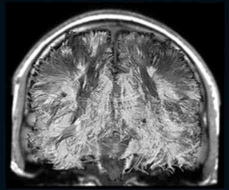
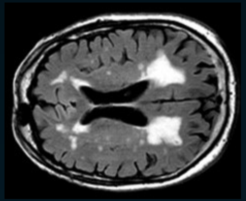
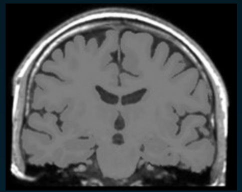
Structural brain imaging correlates of general intelligence in UK Biobank

Cox S.R.[a,b,c,*], Ritchie S.J.[d], Fawns-Ritchie C.[a,b], Tucker-Drob E.M.[e], Deary I.J.[a,b]

- Looked for association between general intelligence and brain volume in >8000 participants (mean 63 years) in UK Biobank study
- Overall correlation of **0.28** between volume and general intelligence

Summary

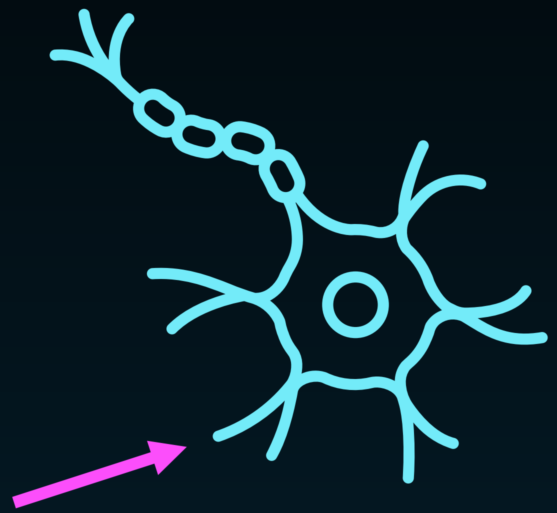
Higher intelligence scores tend to go with having:

- Thicker grey matter on surface of brain 
- Healthier white matter connections overall 
- Fewer white matter scars causing ‘leakage’ 
- A larger brain overall. Why? Answer not fully known – possibly greater numbers of nerve cells 

Single cells and intelligence

What properties of neurons lie at the heart of human intelligence?

- Brain made up of around 100 billion brain cells (building blocks of cognition)
- Each one acts like a small chip – collect, process and pass on information via electrical signals
- Dendrites – long projections to collect signals
- Prediction – larger dendrites will help to initiate electrical signals faster



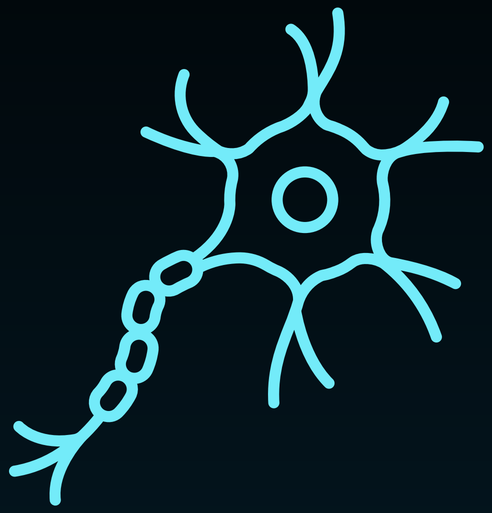
What properties of neurons lie at the heart of human intelligence?

Large and fast human pyramidal neurons associate with intelligence

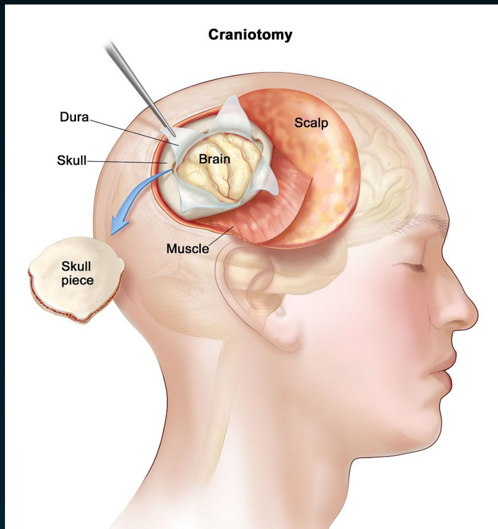
Natalia A Goriounova[1]*, Djai B Heyer[1†], René Wilbers[1†], Matthijs B Verhoog[1], Michele Giugliano[2,3,4], Christophe Verbist[2], Joshua Obermayer[1], Amber Kerkhofs[1], Harriët Smeding[5], Maaike Verberne[5], Sander Idema[6], Johannes C Baayen[6], Anton W Pieneman[1], Christiaan PJ de Kock[1], Martin Klein[7], Huibert D Mansvelder[1]*

eLIFE
elifesciences.org

2018

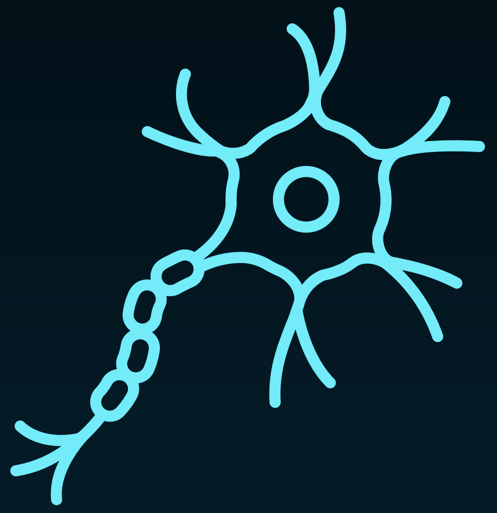


- Wanted to test if smarter brains are equipped with faster and larger brain cells
- Studied 46 people who needed surgery for brain tumours or epilepsy
- Each took an IQ test before the operation

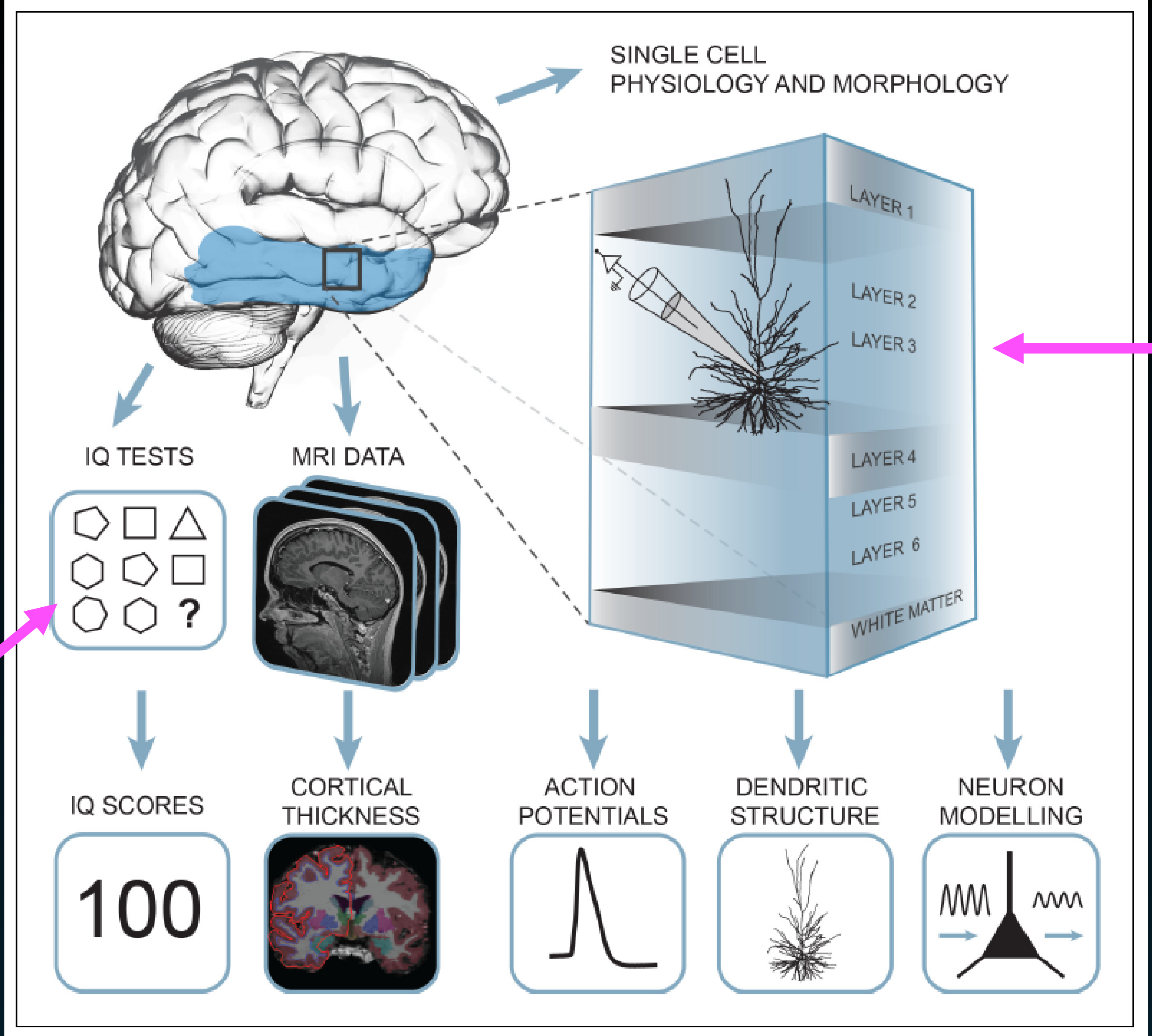


What properties of neurons lie at the heart of human intelligence?

- Vast number of brain cells in human neocortex… so
- even slight change in neuronal efficiency could translate to large differences in mental ability



Goriounova et al. (2018) *eLife*

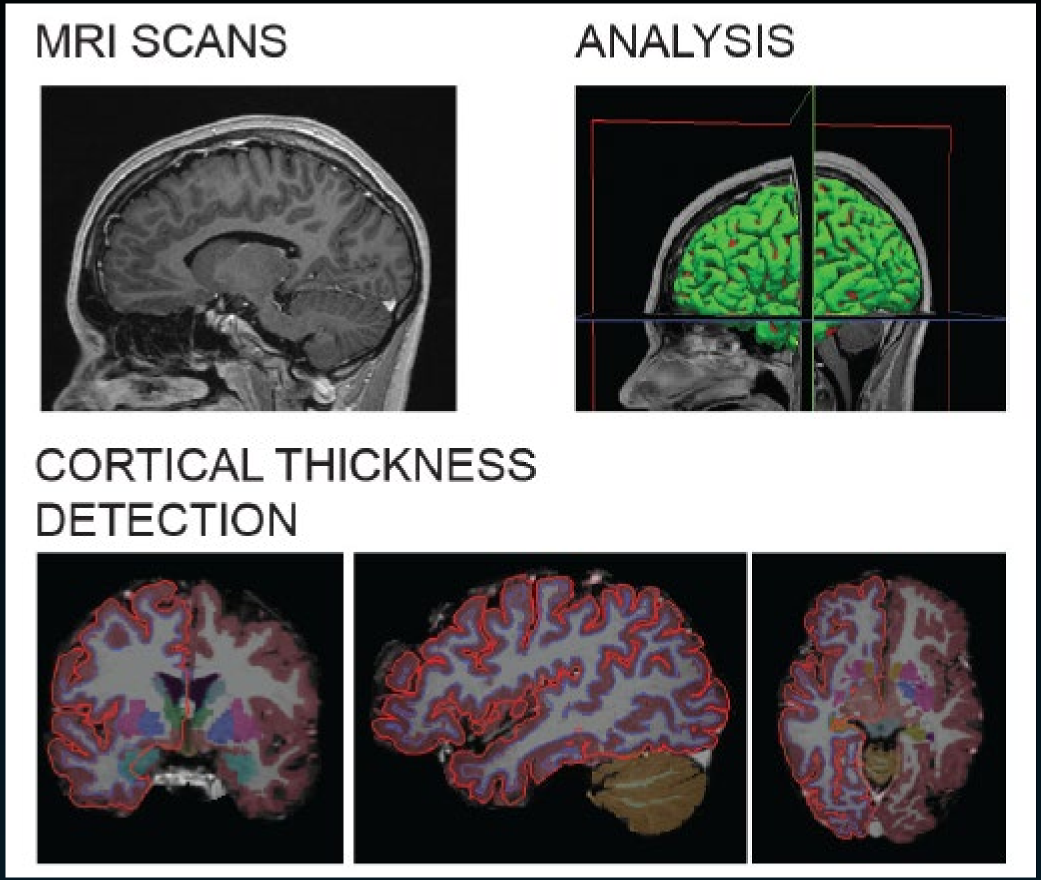


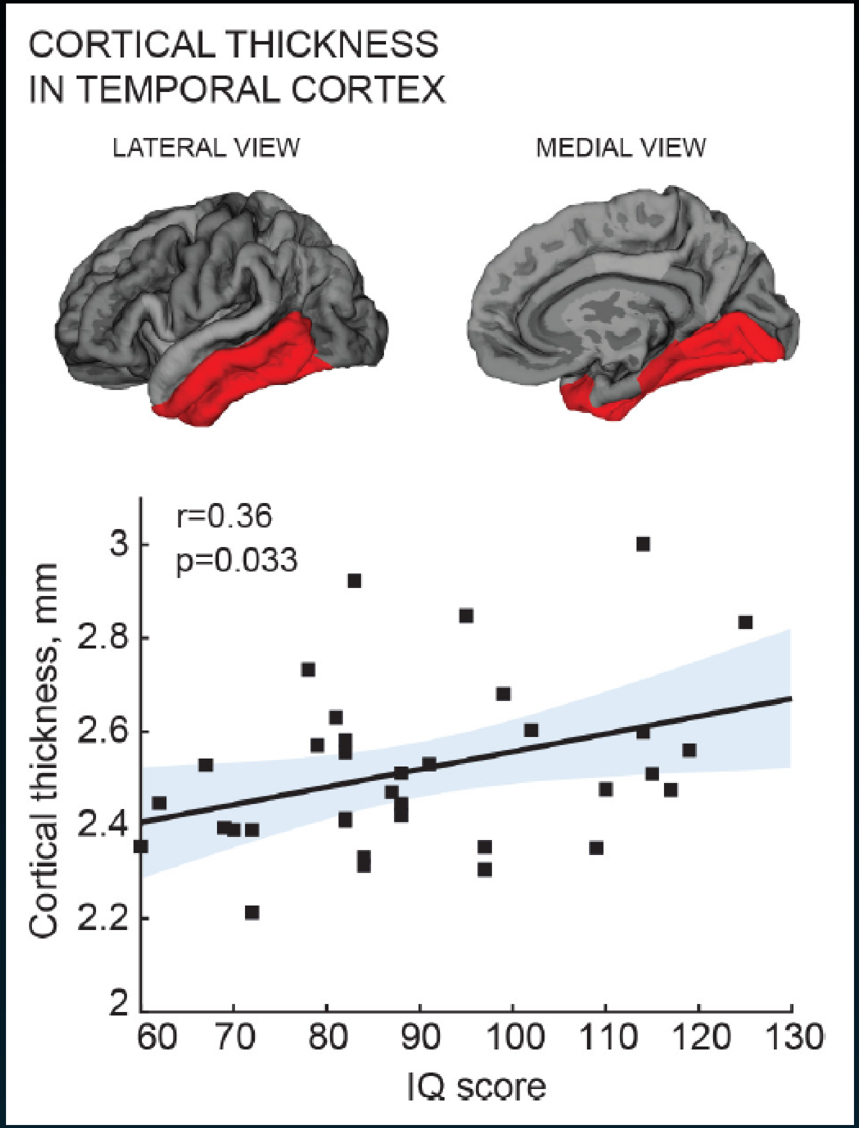
Block of brain tissue was obtained away from the epileptic focus or tumour, and displayed no structural / functional abnormalities

WAIS IQ Test

Goriounova et al. (2018) *eLife*

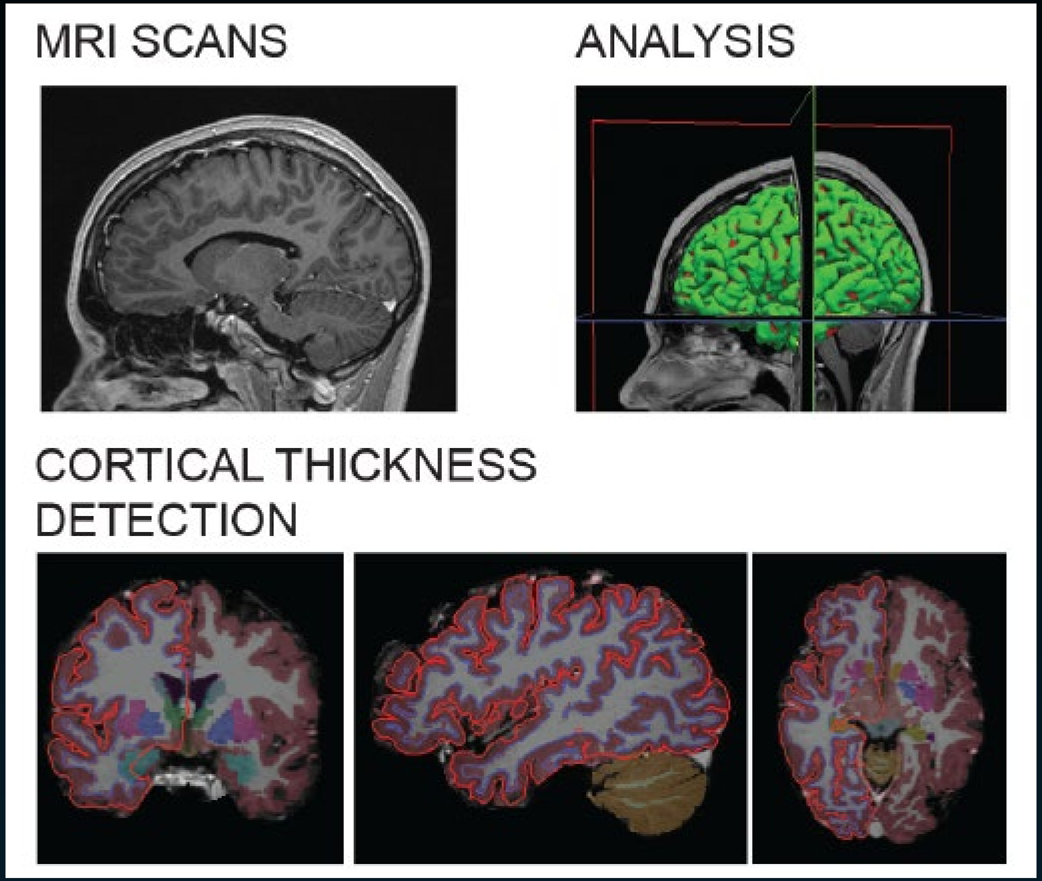
IQ scores positively correlate with cortical thickness of the temporal lobe

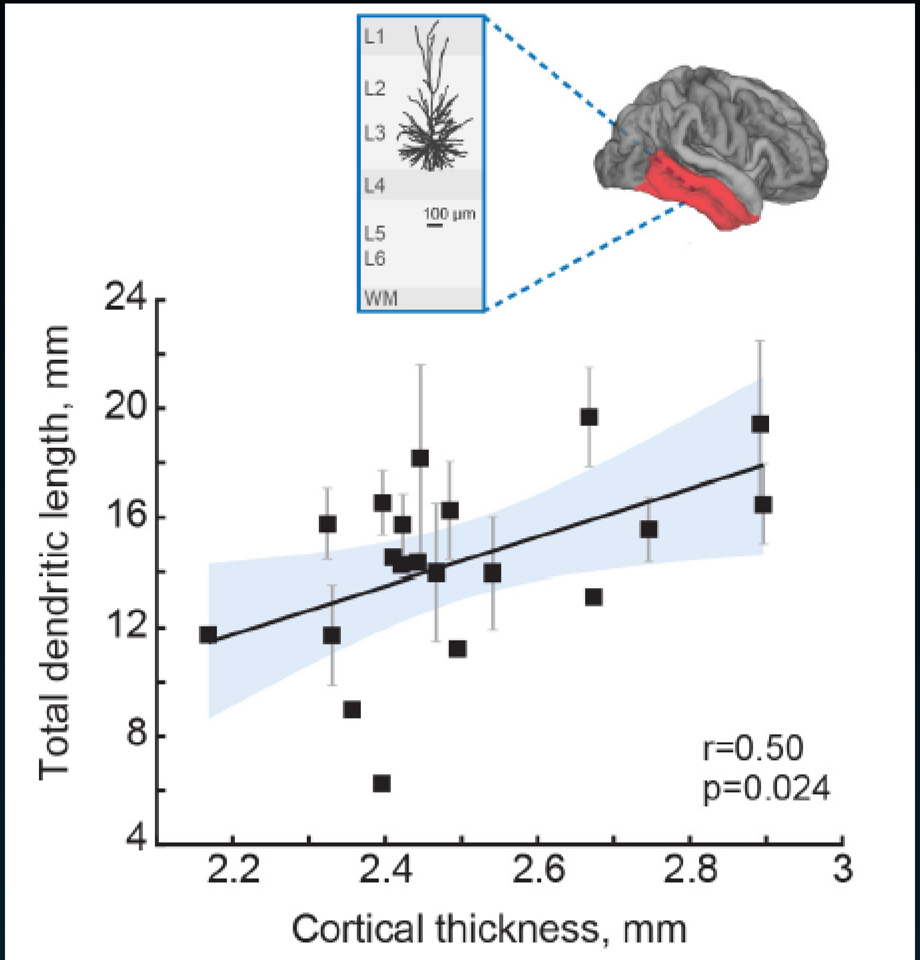




Goriounova et al. (2018) *eLife*

Average total dendritic length in pyramidal cells correlates with cortical thickness in temporal lobe

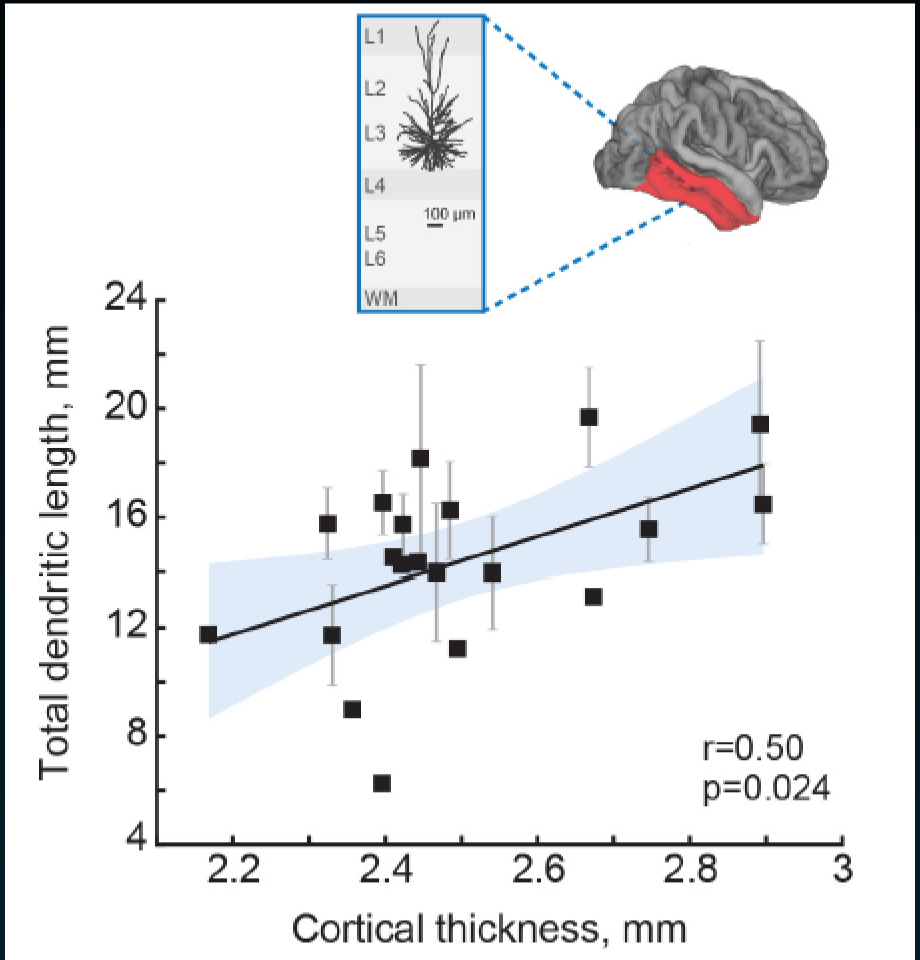




Goriounova et al. (2018) *eLife*

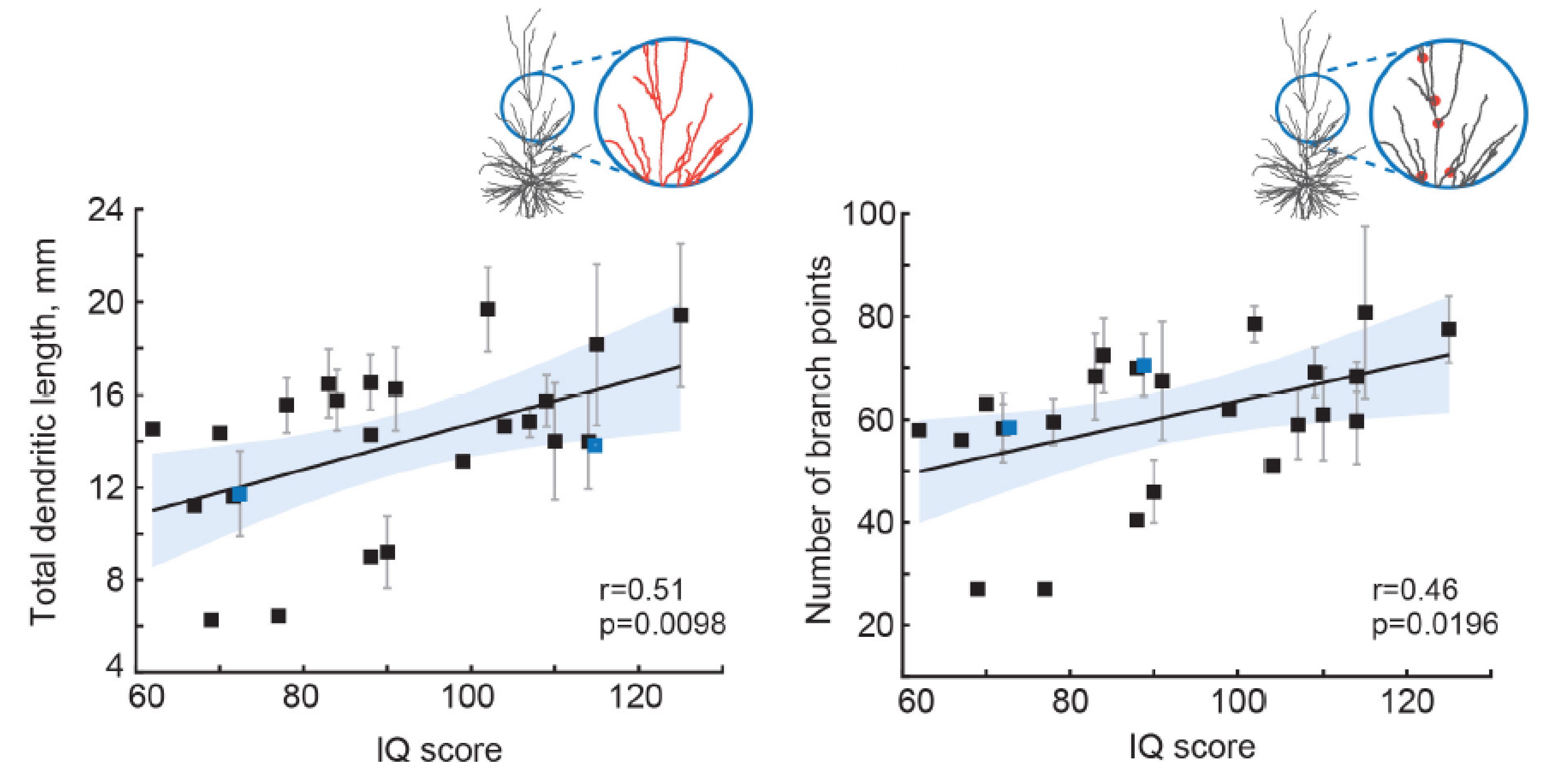
Average total dendritic length in pyramidal cells correlates with cortical thickness in temporal lobe

Indicates that dendritic structure of individual neurons contributes to the overall thickness of cortex



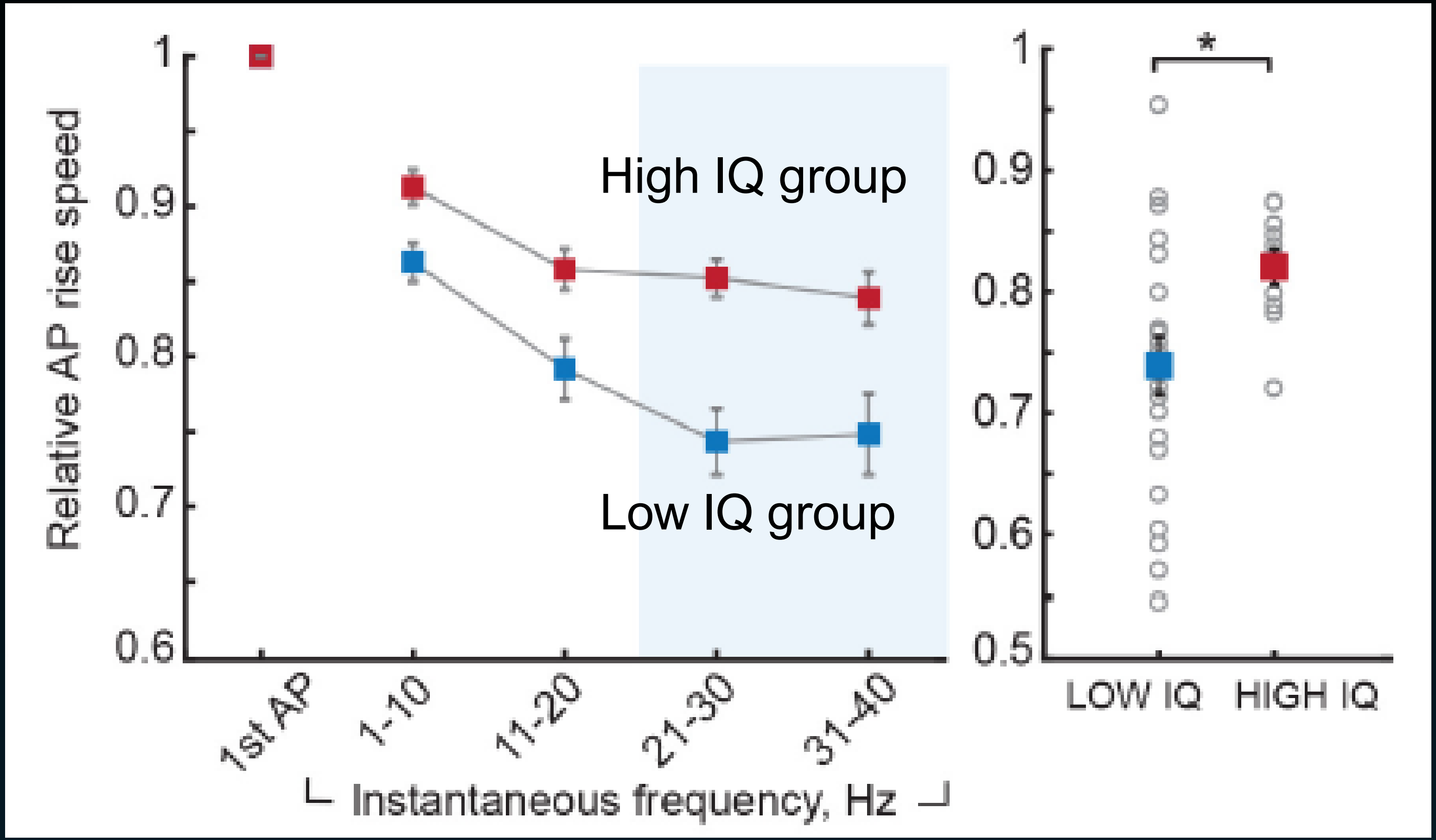
Goriounova et al. (2018) *eLife*

Average total length and number of dendritic branches in pyramidal cells correlates with IQ scores



Goriounova et al. (2018) *eLife*

Higher IQ scores are accompanied by faster action potentials

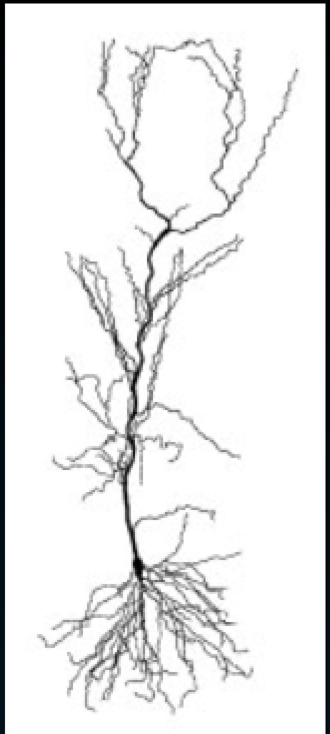


Action potentials slowed down stronger in individuals with lower IQ scores compared to higher IQ scores

Goriounova et al. (2018) *eLife*

Summary

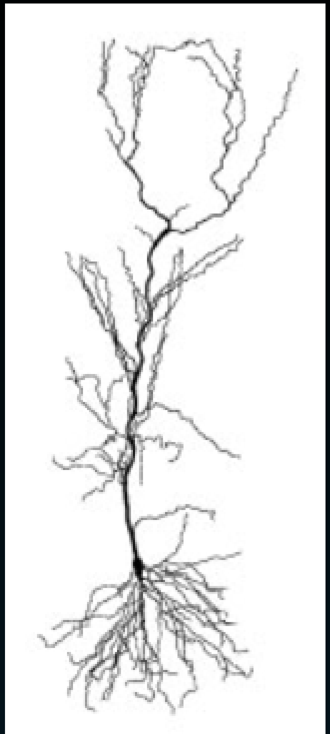
- Pyramidal cells – abundant in cerebral cortex of virtually every mammal studied
- Primarily found in brain structures associated with advanced cognitive functions, e.g. intelligence
- Pyramidal cells are the primary excitation units in cortex - integrators and accumulators of synaptic information



Goriounova et al. (2018) *eLife*

Summary

Goriounova et al. (2018) provide biological cellular evidence that individuals with a higher IQ possess larger / more complex dendrites and are able to sustain faster action potentials



Goriounova et al. (2018) *eLife*

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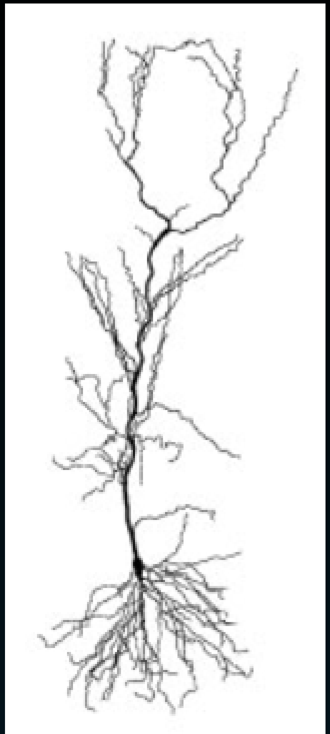


In conditions of high cognitive demand, like during an IQ test, larger and faster neurons can help to process information quickly and efficiently

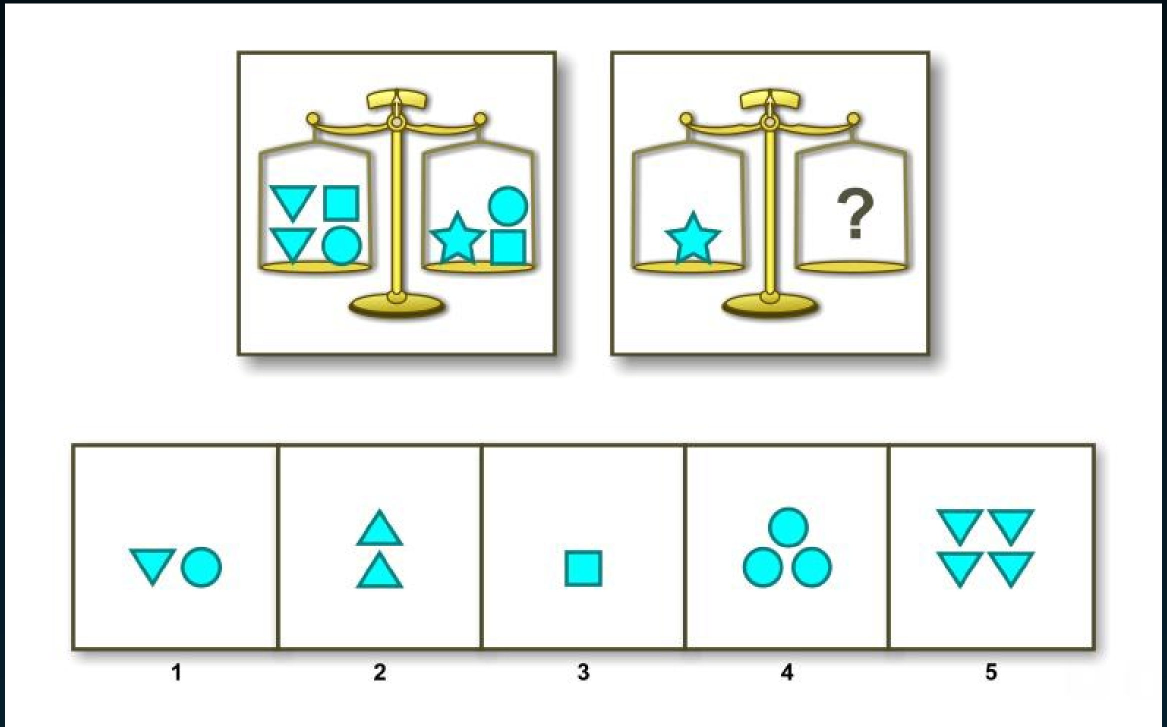
Goriounova et al. (2018) *eLife*

Brain regions and Intelligence

Where is the seat of intelligence in the brain?

- Widely accepted theory that working memory is crucial for fluid intelligence (g_f), that is the ability to use inductive, deductive, and quantitative reasoning to solve novel, "on-the-spot" problems

- Fluid intelligence (g_f) is distinct from crystallized intelligence (g_c) which refers to overlearned skills and prior knowledge (vocabulary, general knowledge facts etc)



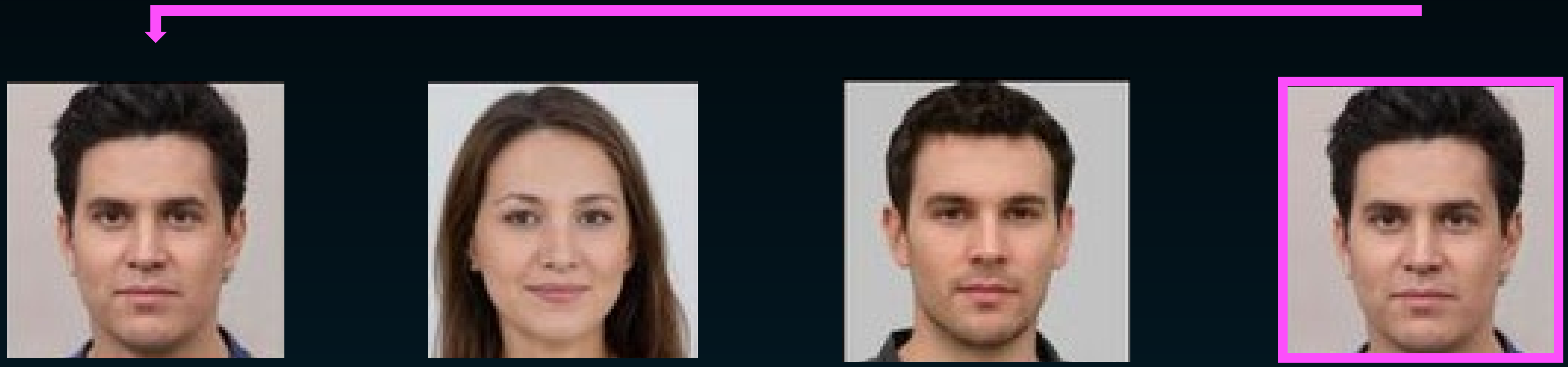
Where is the seat of intelligence in the brain?

- Gray et al (2003) tested whether fluid intelligence (g_f) was mediated by neural mechanisms supporting working memory
- 48 healthy participants
- Raven's Advanced Progressive Matrices, widely used measure of g_f, was administered before fMRI session
- They probed individual differences in general fluid intelligence and looked to see if these differences corresponded to differences in brain activity

Gray et al. (2003) *Nature Neuroscience*

Where is the seat of intelligence in the brain?

- Measured performance on the ‘n back task’ – a test of working memory. Here they used a three-back task:



Match

Non-match

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Gray et al. (2003) *Nature Neuroscience*

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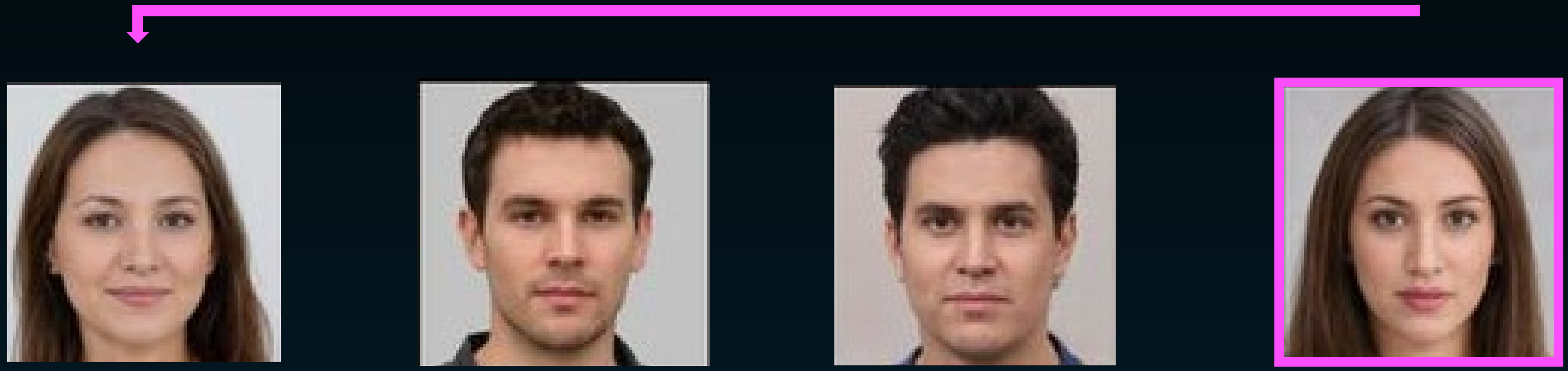

Match

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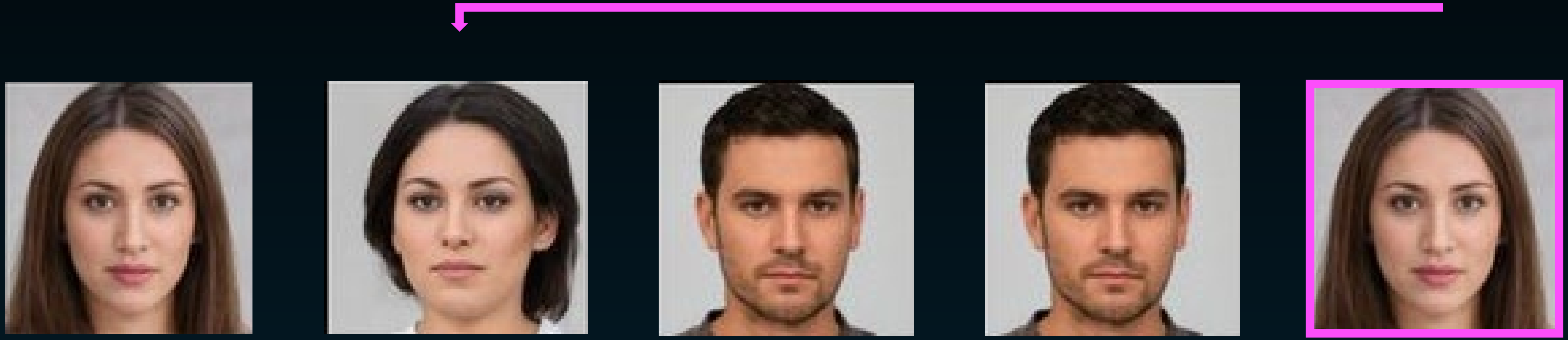
Gray et al. (2003) *Nature Neuroscience*

Where is the seat of intelligence in the brain?

Another high attentional demand was included on non-target trials. Lures were included - faces appear before the target but in the wrong place:



Match

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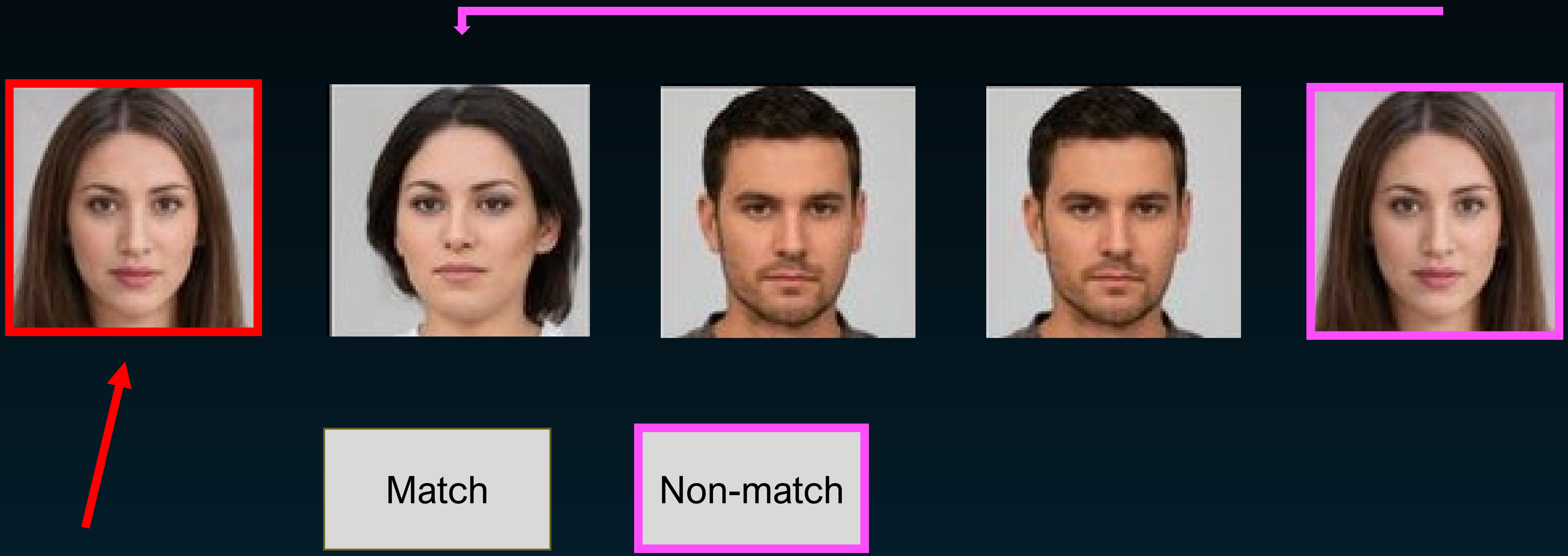


Confusing Lure

Button press

Gray et al. (2003) *Nature Neuroscience*

Where is the seat of intelligence in the brain?

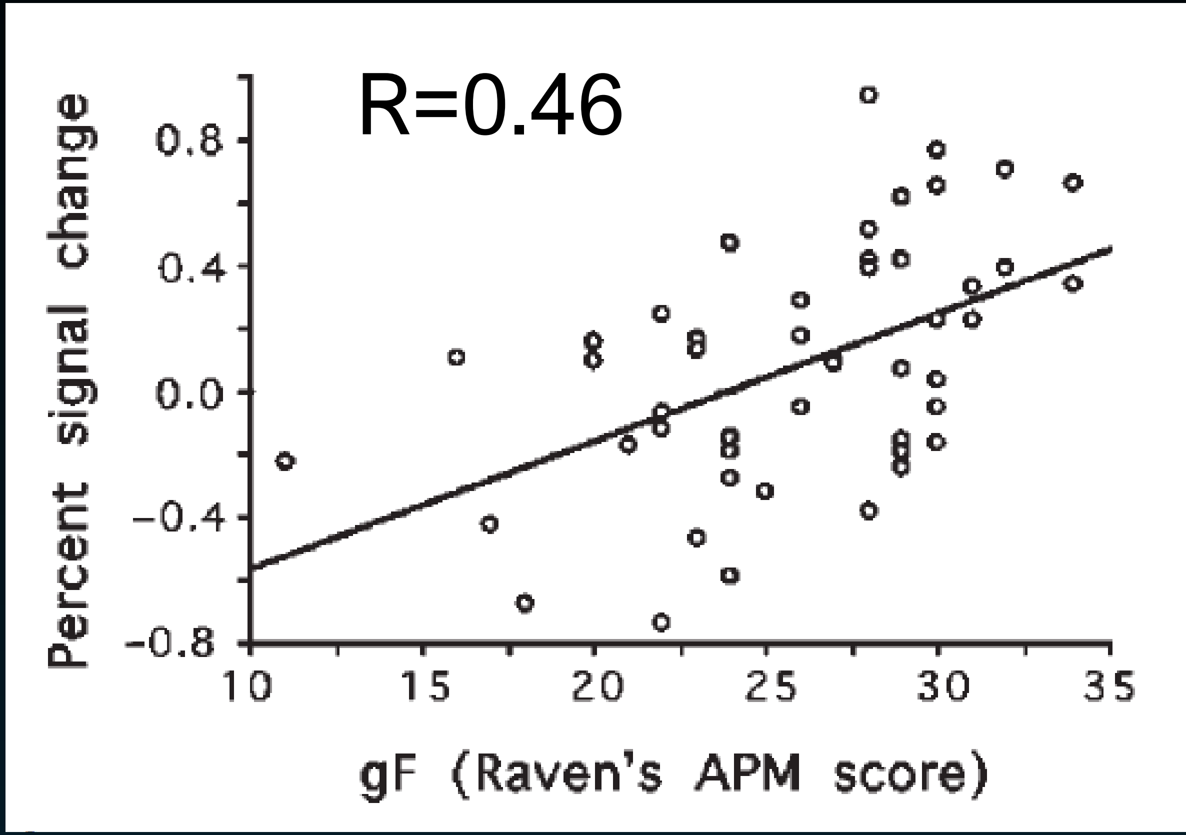
Before considering brain activity, consider the performance correlation between IQ score (Raven's Advanced Matrices) and working memory (n-back)

There was a positive correlation (**0.36**) between accuracy on 'Lure' trials (higher interference working memory) and fluid intelligence (Raven's)

Gray et al. (2003) *Nature Neuroscience*

Where is the seat of intelligence in the brain?

Left lateral PFC

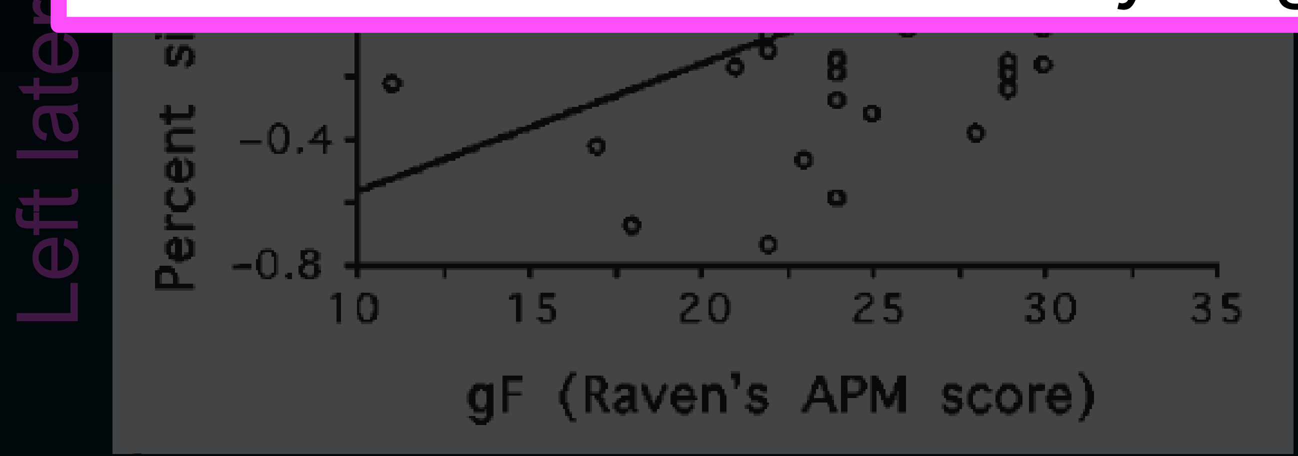


fMRI showed that activity in left lateral Prefrontal cortex in correct 'Lure' trials (Y axis) correlates with fluid intelligence on Raven's Matrices (X axis)

Gray et al. (2003) *Nature Neuroscience*

Where is the seat of intelligence in the brain?

Conclusion: Participants scoring higher on Raven's Matrices Tests (fluid intelligence) were more accurate in a challenging working memory task during which left Prefrontal cortex was heavily engaged



with fluid intelligence on Raven's Matrices (X axis)

Gray et al. (2003) *Nature Neuroscience*

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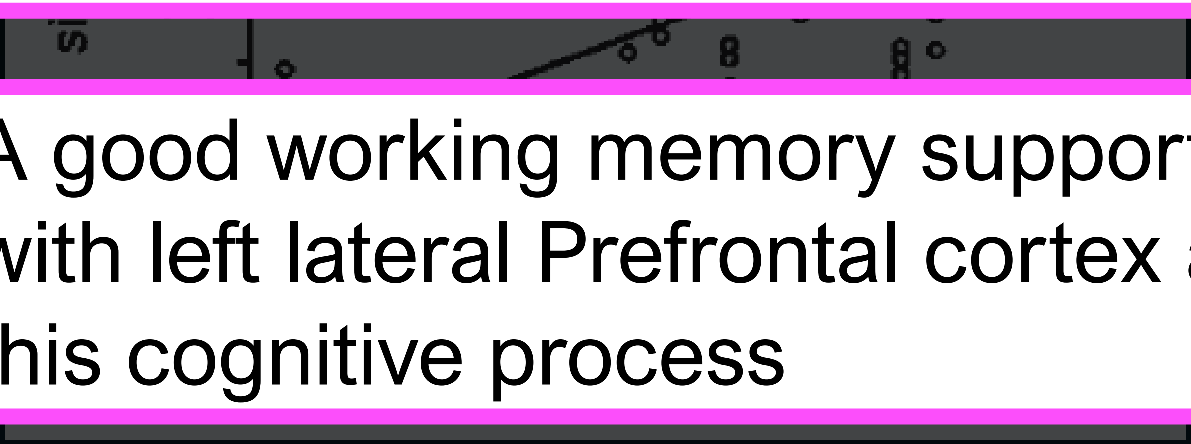


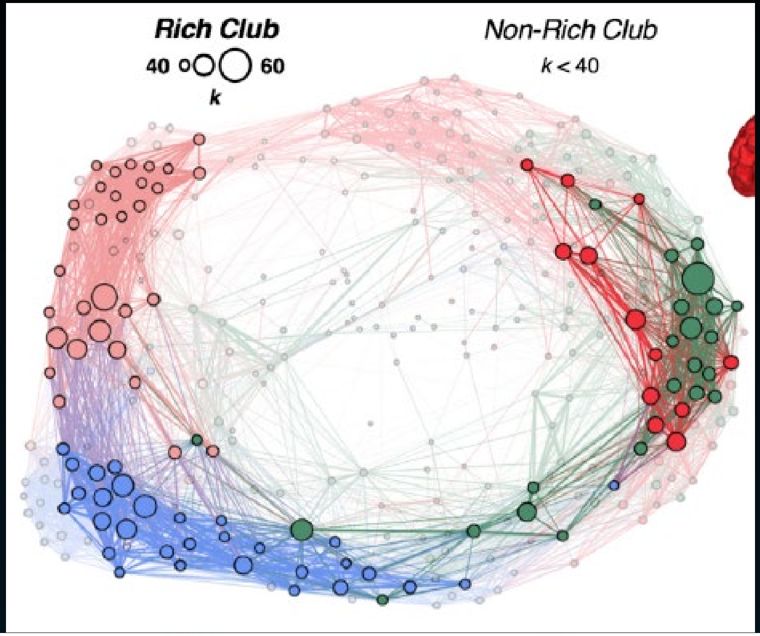
A good working memory supports fluid intelligence (g_f) with left lateral Prefrontal cortex activity partly mediating this cognitive process

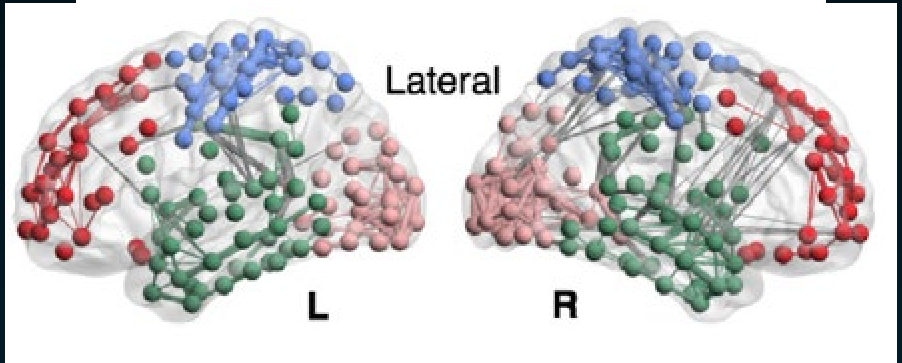
Gray et al. (2003) *Nature Neuroscience*

Brain connectivity and intelligence

Are smarter brains more connected than others?

- Mapping the connections in the brain – known as the ***connectome***
- 296 healthy adults / adolescents (14-24 years) and validated with a second cohort of 124 volunteers
- Used multimodal MRI to build ‘morphometric similarity networks (MSNs)’ to show how well connected an individual’s brain regions are
- Built a map showing how well connected the ‘hubs’ were





Seidlitz et al. (2018) *Neuron*

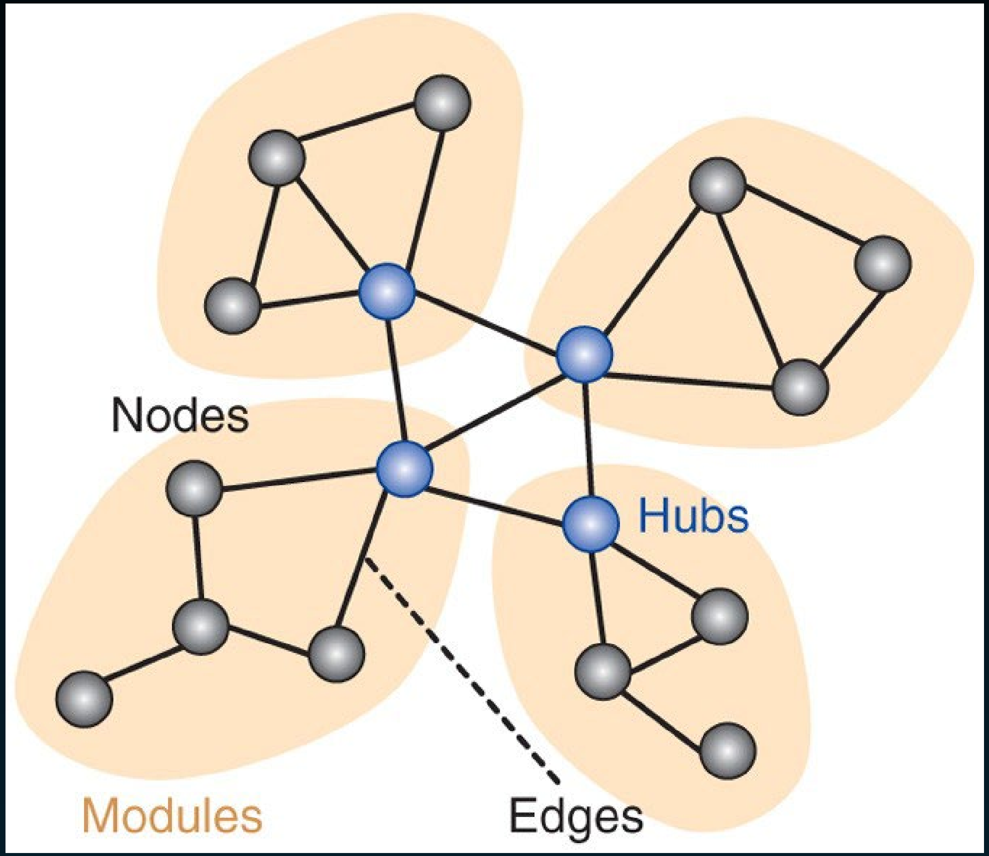
Are smarter brains more connected than others?

Investigated relationship between interindividual differences in brain network topology and interindividual differences in general intelligence

Predicted specifically that individual differences in IQ should be related to individual differences in nodal degree

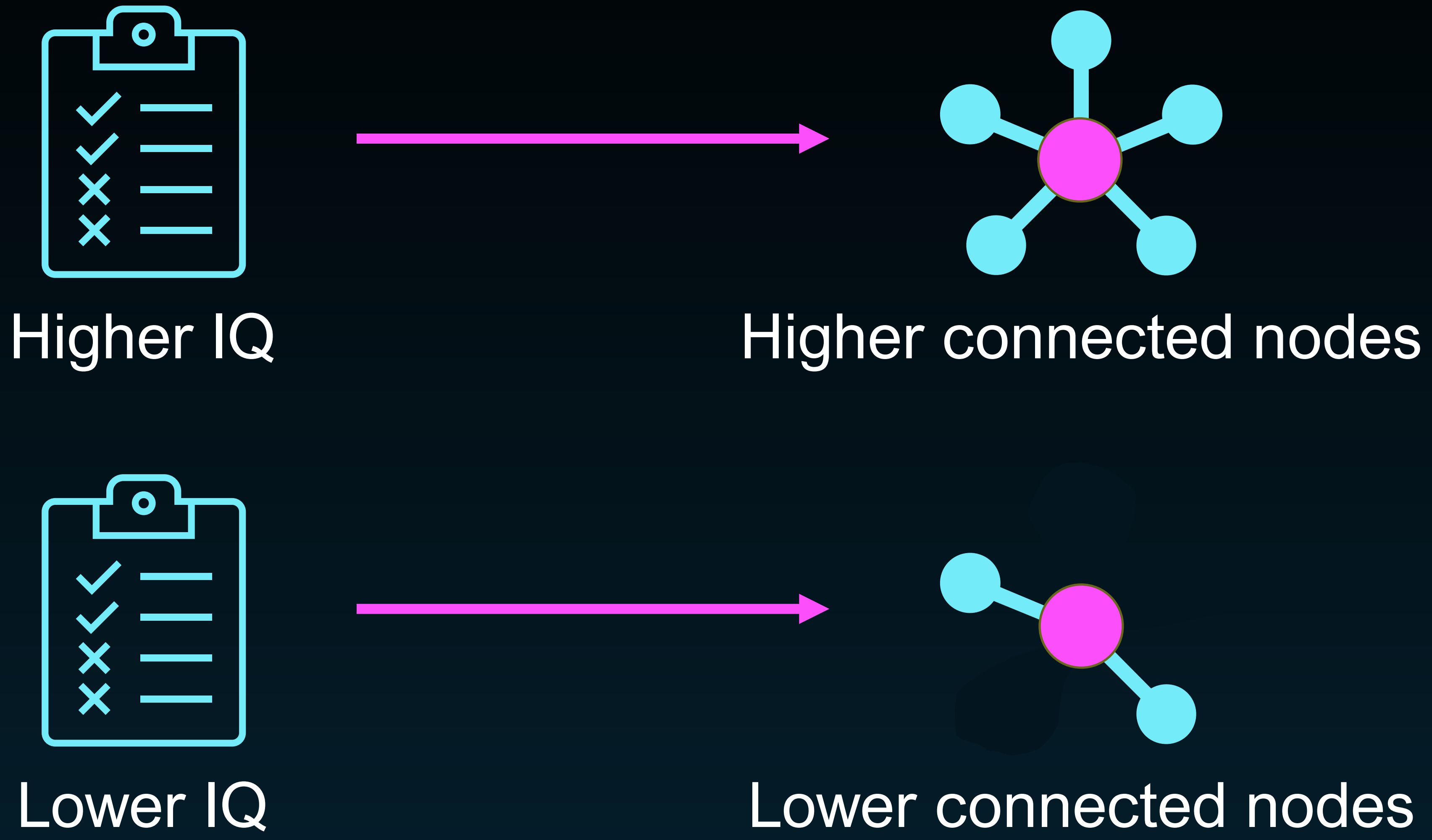
'Hubs' – high number of pyramidal neurons that can simultaneously capture multi dimension info during higher cognitive functions

Hubs can coordinate activity of large populations of globally distributed neurons

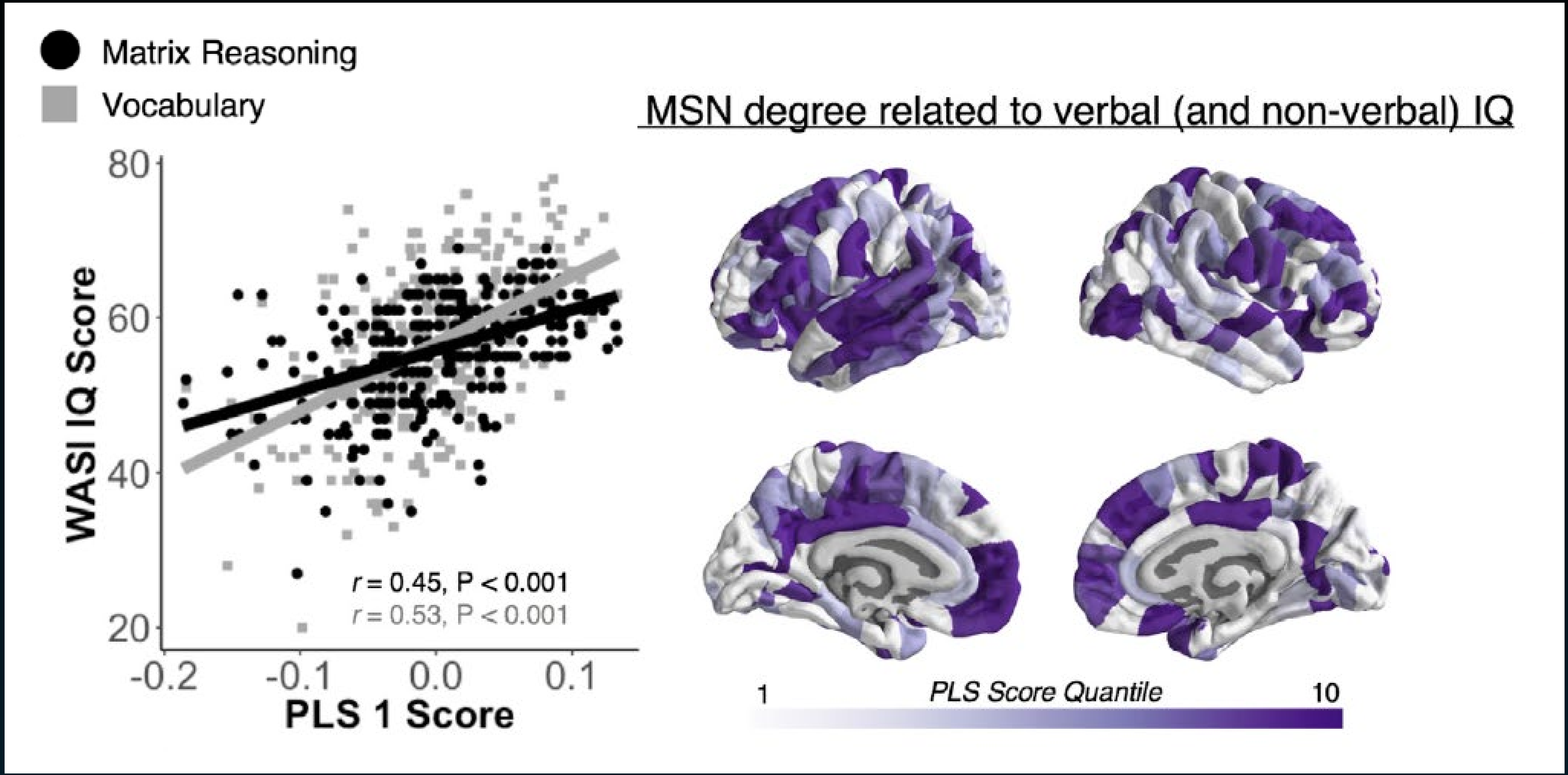


Seidlitz et al. (2018) *Neuron*

Are smarter brains more connected than others?



Are smarter brains more connected than others?



High degree (highly connected) hubs identified in left frontal and temporal cortex, which were highly predictive of higher IQ score

Are smarter brains more connected than others?

We saw a clear link between the 'hubbiness' of higher-order brain regions – in other words, how densely connected they were to the rest of the network – and an individual's IQ. This makes sense if you think of the hubs as enabling the flow of information around the brain – the stronger the connections, the better the brain is at processing information."

Jakob Seidlitz

Overall Summary

SUMMARY

Higher intelligence scores correlate with:

- Larger whole-brain volume
- Thicker grey matter on brain surface (cortical thickness)
- Larger / more complex dendrites and faster action potentials
- Good working memory (partly mediated by PFctx)
- Highly connected higher-order brain networks